

CARBENTRA / CM-S16-EVT-B INTEGRATED ELECTRICAL SOURCE

101 main-board parts + 4 remote-head parts + 4 off-board assemblies; harness models are not PCB parts

Hierarchy is functional grouping. Shared global net names are real electrical connections.

CANDIDATE ONLY: no fabrication, energization, safety or certification release

AUXILIARY FUSE: 1

1 controller

PS101 IRM-10-5: AP63203 3.3 V rail
ESP32-C3 + low-side relay driver
K101: HOT_LOAD to HOT_SWITCHED
Coil feed = hardware-gated COIL_BV
OUTPUT_PRESENT_N enters U101.15

File: Integrated_controller.kicad_sch

2 meter

RS201 Kelvin shunt + voltage divider
ATW90E26 AFE + reinforced SPI isolation
R05CT Isolated supply for HOT_3V3
Powered ASV clock / CX201 shown at right
HOT_GND is live line reference

File: Integrated_meter.kicad_sch

POWERED CLOCK: 1

3 feedback

Four 33k resistors + ACPL-K376 detector
Senses HOT_SWITCHED relative to HOT_N
OUTPUT_PRESENT_N returns to the MCU
Active-low 100/120 Hz pulses
No proof of dead output or contact position

File: Integrated_feedback.kicad_sch

4 thermal

Remote head: U401 / R401 / C401 / J404
TMP302 + supervisor + trip-dominant latch
SW401 physical rearm only; no MCU rearm
Q401 gates coil feed; TPA01 / TPA02 optional
Backside coupling / Insulation needs qualification

File: Integrated_thermal.kicad_sch

5 assembly protection

Plug L -> F601 main fuse -> TF601 thermal link -> protected main PCB input
Socket output follows K101; PE is direct and never enters PCB or relay

File: Integrated_assembly.kicad_sch

#FLG1
EXTERNAL / PASSIVE RAIL

+3V3_AUX

#FLG5
EXTERNAL / PASSIVE RAIL

THC3V3_HEAD

#FLG2
EXTERNAL / PASSIVE RAIL

HOT_N

#FLG6
EXTERNAL / PASSIVE RAIL

THC6V6_HEAD

#FLG3
EXTERNAL / PASSIVE RAIL

HOT_AUX_FUSED

#FLG7
NC LIVE-POTENTIAL NET ANCHOR

HOT_NC

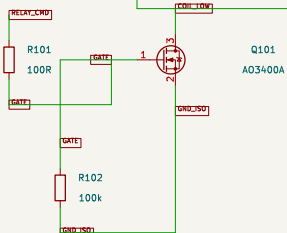
HOT_GND is LINE POTENTIAL. PE is external, continuous and never switched.

Remote head: component face rear; smooth GND copper backside to qualified Insulating pad. Thermal lag/Insulation HOLD.

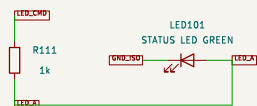
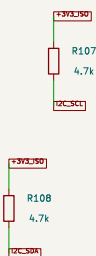
04 EDGE CONTROLLER



05 LOCAL CONTROLS



06 BOARD TEMPERATURE

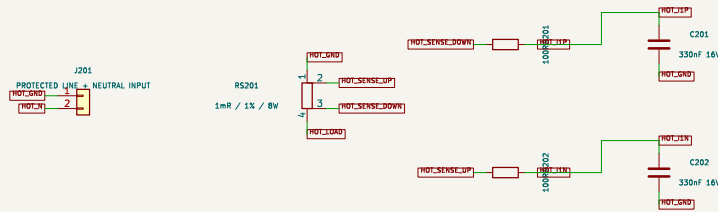


CARBONMIRROR REV B - METER - DEVELOPMENT / ENERGIZATION HOLD

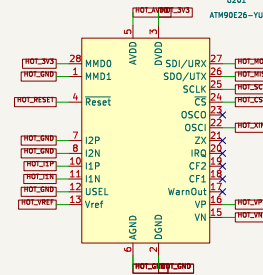
HOT_* = LIVE LINE REFERENCED. Never connect hot ground to PE, USB, isolated ground or accessible metal.

Protected line -> Kelvin shunt -> relay COM. Voltage samples Input L-N: separate ACPL-K376 senses switched output.

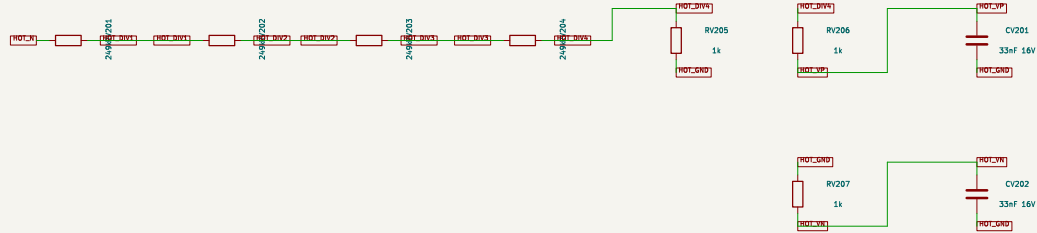
01 LOAD SHUNT + MATCHED RC



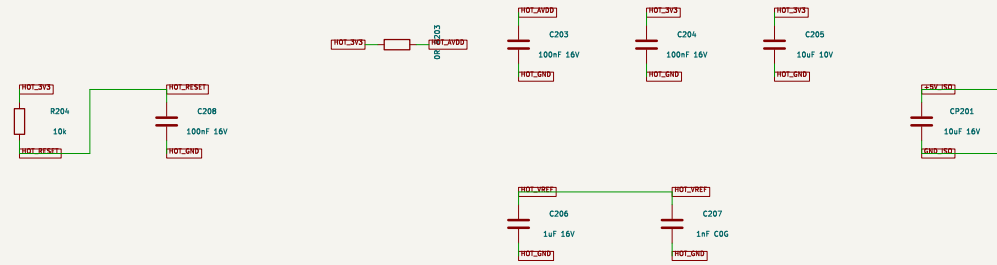
03 ATM90E26 AFE



02 VOLTAGE DIVIDER + FILTER



06 RESET / DECOUPLING; POWERED CLOCK ON OVERVIEW



04 REINFORCE



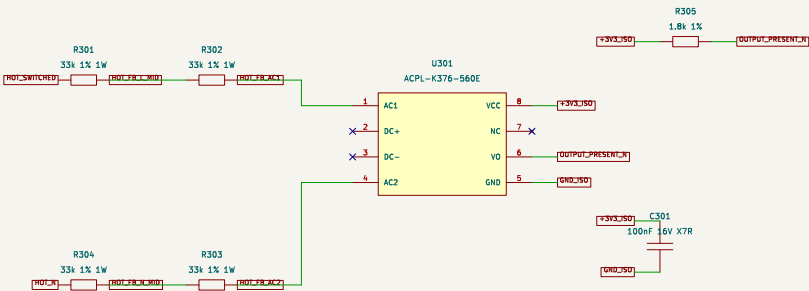
05 ISOLATED P



REV B OUTPUT PRESENCE / DEVELOPMENT ONLY

198-242 VAC OPERATING PROPOSAL; 264 VAC ANALYSIS HEADROOM ONLY; QUALIFICATION HOLD

MAINS INPUT: tap the ACTUAL socket contacts after all switched poles



ISOLATED 3.3 V; active-low pulses, 100/120 Hz when AC is present

DC+ / DC- / NC Intentionally open. Do not bridge either isolation domain.

No safety Isolation proof, no proof of absence of voltage, no physical contact-position feedback.

132k total series resistance; built-in bridge/threshold controller inside U1.

Use a separate hardware thermal Interlock AND a qualified series thermal cutoff.

Sheet: /3 feedback/
File: integrated_feedback.kicad_sch

Title: CARBENTRA EVT-B: feedback

Size: A4
KiCad E.D.A. 9.0.2

Date:

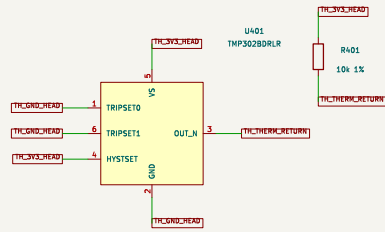
Rev: EVT-B HOLD
Id: 4/6

REV B HARDWARE THERMAL TRIP + LOCAL PHYSICAL REARM ONLY

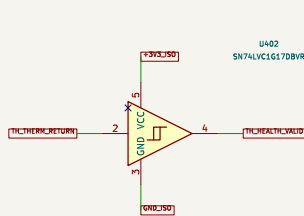
DEVELOPMENT CANDIDATE – NO ENERGIZATION OR FABRICATION RELEASE

REMOTE HEAD: U401 / R401 / C401 / J404 only. W401/W402 are cable models, not PCB parts.

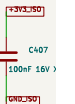
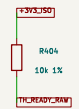
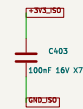
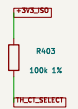
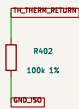
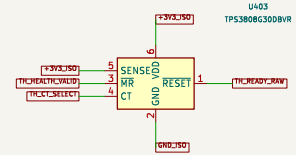
01 REMOTE TEMPERATURE HEAD



02 WIRE-OPEN FAIL-LOW

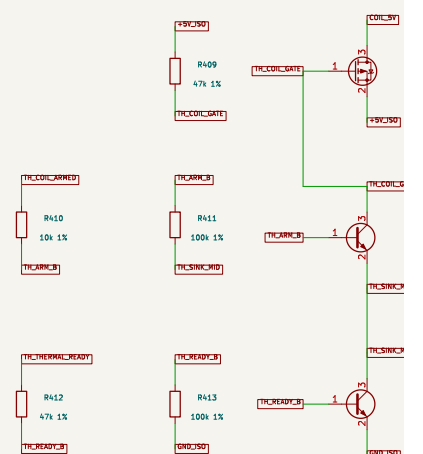
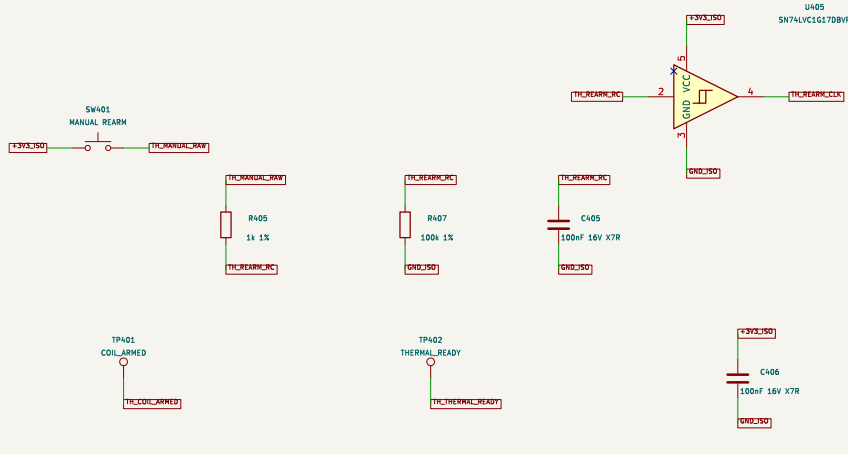


03 POWER + THERMAL SETTLING



06 SERIES COIL FEED PERMISSION

05 FRESH REARM EDGE (DO NOT GATE CLOCK WITH READY)

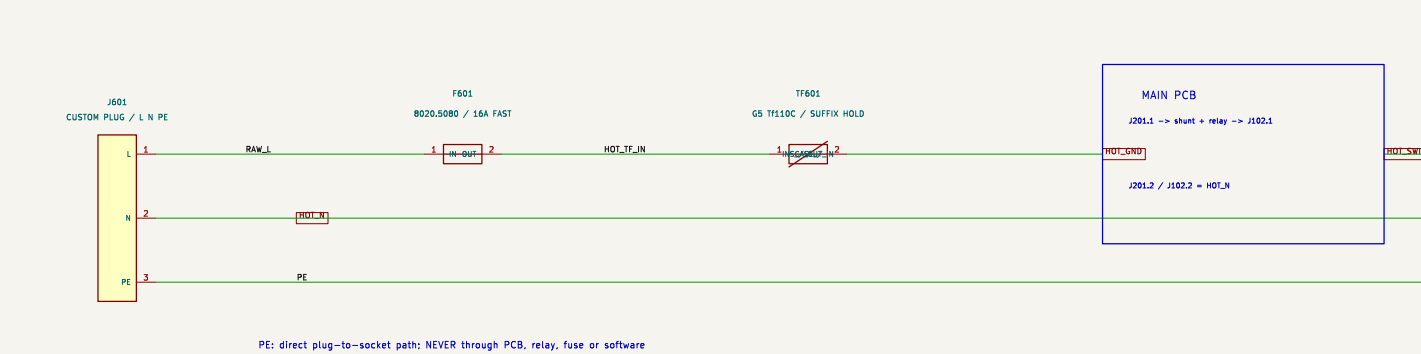


COIL_5V feeds K101.A1 and D101 cathode; no ungated coil-supply bypass is permitted.

A coil interlock cannot interrupt welded load contacts. Separate qualified line-series thermal cutoff remains required.

R10 ASSEMBLY PROTECTION / OFF-BOARD INTERNAL PARTS

CANDIDATE ONLY: this sheet closes the drawn assembly path, not qualification or certification



F601 = MAIN 16 A fast-acting fuse. F501 = separate PCB auxiliary 1 A time-delay fuse.

TF601 = distinct one-shot MICROTEMP G5 thermal link, provisional TF110C; order suffix and mounting HOLD.

SHF16A1.5kA@250VAC cos(phi)0.7-0.8. Typical melting I2t1760A2s is NOT total clearing or a downstream energy bound.

Upstream breaker, prospective fault current, fuse I2t/selectivity, hot-condition derating and wiring coordination are UNVERIFIED.

NO ACTIVE CURRENT LIMIT: a fuse rating is not a continuous-current clamp or a guaranteed socket overload threshold.

Custom contacts + 2x CQP8040.0003 clips;45x16x2 carrier / Cu collectors; fit accepted, heating/insulation HOLD.

Assembly contact numbers identify L/N/PE wiring. SHF8020.5080 fuse6.3x32mm replaces216/0GN for higher clip margin.

The main-PCB box references real circuits on sheets 1-4; it is not an electrical bypass between HOT_GND and HOT_SWITCHED.

PE continuity, polarity, touch protection, contact heating and cold-weld disconnection still require physical verification.

F601 / TF601 / J601 / J602 are on_board=false. Main PCB part count remains 101.